

Interactive Medical Signal Analysis Workbench

A Scilab-Based GUI for ECG Visualization, Signal Processing,
and Educational Biomedical Analysis

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1 Project Title

Interactive Medical Signal Analysis Workbench

A Scilab-Based Graphical Application for ECG Visualization, Digital Signal Processing, and Educational Biomedical Signal Analysis

2 Project Description

2.1 Background

Electrocardiography (ECG) is one of the most widely used non-invasive diagnostic techniques for monitoring the electrical activity of the human heart. Engineers and students studying biomedical signal processing frequently require tools to visualize ECG waveforms, preprocess signals, analyze frequency components, detect important cardiac events, and evaluate statistical characteristics.

While several software platforms provide ECG analysis capabilities, many are either commercial, heavily script-based, or intended primarily for clinical diagnosis rather than educational purposes. Consequently, beginners often spend significant time learning programming syntax instead of understanding the underlying biomedical signal processing concepts.

2.2 Motivation

The motivation behind this project was to develop an integrated, open-source graphical application that enables users to perform ECG visualization, digital signal processing, statistical analysis, and data export through an intuitive graphical interface.

Rather than executing multiple independent scripts, users can complete the entire analysis workflow from a single application.

2.3 Objective

The Interactive Medical Signal Analysis Workbench has been developed as an educational platform for learning biomedical signal processing using Scilab.

The application enables users to:

- Load ECG datasets from CSV files.
- Visualize heartbeat waveforms.

- Navigate between individual beats.
- Apply digital filters.
- Perform FFT analysis.
- Detect ECG peaks.
- Compute statistical parameters.
- Export processed ECG signals.
- Save generated plots.

2.4 Intended Users

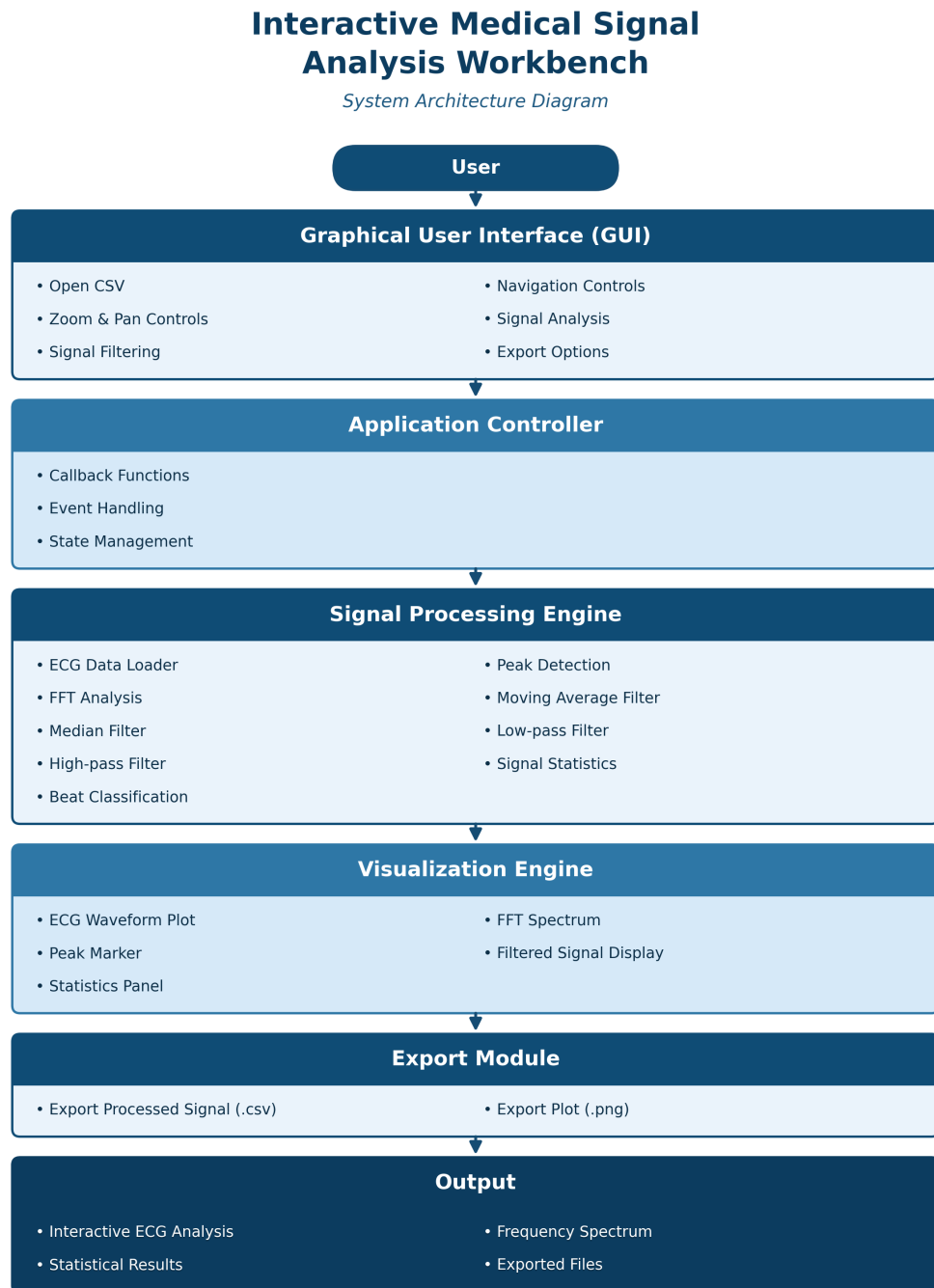
- Biomedical Engineering students
- Electronics Engineering students
- Signal Processing learners
- Researchers
- Laboratory instructors

3 Software Requirements

Table 1: Software Requirements

Component	Requirement
Operating System	Windows 10/11 or Ubuntu Linux
Development Platform	Scilab 2025
Processor	Intel Core i3 or Higher
RAM	Minimum 4 GB (8 GB Recommended)
Storage	200 MB Free Space
Display Resolution	1366×768 or Higher
Dataset Format	CSV
Additional Software	None

4 System Architecture



Developed using Scilab GUI Components and Signal Processing Functions.

Figure 1. System architecture of the Interactive Medical Signal Analysis Workbench illustrating the interaction between the graphical user interface, application controller, signal processing modules, visualization engine, and export module.

5 Toolboxes Used

The project has been developed entirely using native Scilab functionality without requiring external toolboxes.

The application makes use of:

- GUI Development (figure, uicontrol, newaxes)
- Matrix Operations
- Numerical Computation
- FFT
- Digital Filtering
- CSV File Handling
- Scientific Plotting

No third-party libraries or proprietary software are required.

6 Steps to Run the Application

1. Install Scilab 2025.
2. Download the project.
3. Open the project folder.
4. Set the current directory.
5. Execute:

```
exec("main.sce",-1);
```

6. The GUI launches automatically.
7. Click Open CSV.
8. Load an ECG dataset.
9. Perform analysis using the available tools.

7 GUI Explanation

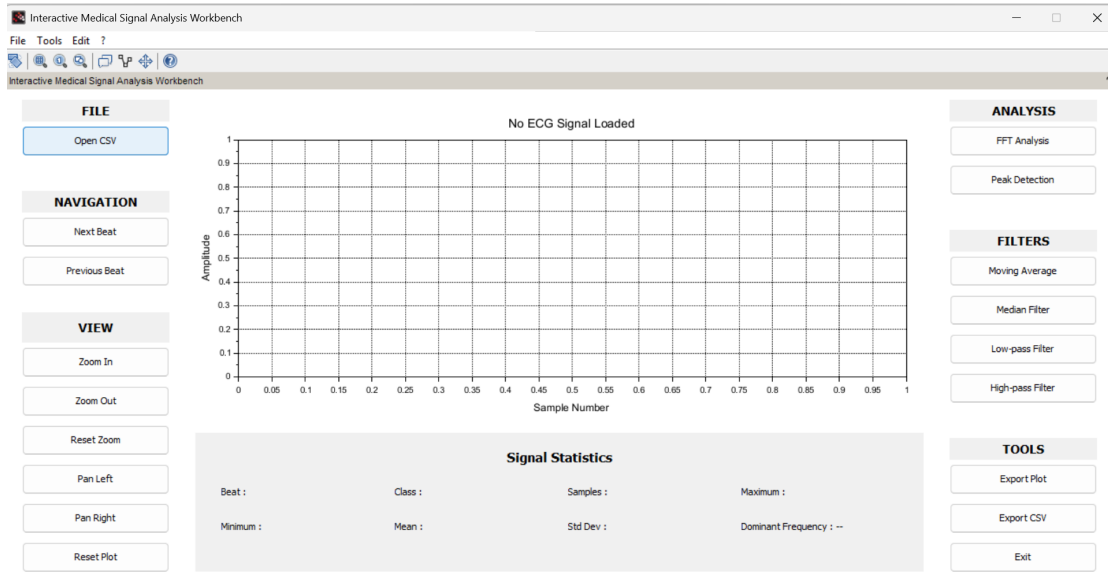


Figure 1. Main interface of the Interactive Medical Signal Analysis Workbench showing the ECG visualization area, control panels, and real-time statistics panel.

7.1 Left Control Panel

- Open CSV – Loads an ECG dataset from a CSV file.
- Next Beat – Displays the next heartbeat in the dataset.
- Previous Beat – Displays the previous heartbeat.
- Zoom In – Magnifies the ECG waveform for detailed analysis.
- Zoom Out – Reduces the zoom level to view more of the signal.
- Reset Zoom – Restores the default viewing scale.
- Pan Left – Moves the displayed waveform towards the left.
- Pan Right – Moves the displayed waveform towards the right.
- Reset Plot – Restores the original unprocessed ECG signal.

7.2 Central Plot Area

- ECG waveform visualization – Displays the selected ECG heartbeat graphically.

- Dynamic updates – Refreshes automatically after each analysis or filtering operation.
- Zooming – Allows detailed inspection of specific waveform regions.
- Panning – Enables horizontal navigation across the ECG signal.
- Peak highlighting – Marks the detected R-peak on the ECG waveform.

7.3 Right Control Panel

- FFT Analysis – Displays the frequency spectrum of the ECG signal.
- Peak Detection – Detects and highlights the dominant ECG peak.
- Moving Average Filter – Smooths the signal by reducing random noise.
- Median Filter – Removes impulse noise while preserving waveform features.
- Low-pass Filter – Attenuates high-frequency noise components.
- High-pass Filter – Removes baseline drift and low-frequency interference.
- Export CSV – Saves the processed ECG signal as a CSV file.
- Export Plot – Saves the current graph as an image.
- Exit – Closes the application safely.

7.4 Statistics Panel

Displays

- Beat Number – Indicates the currently selected heartbeat.
- Beat Class – Shows the classification label of the heartbeat.
- Number of Samples – Displays the total samples in the signal.
- Maximum Value – Shows the highest signal amplitude.
- Minimum Value – Shows the lowest signal amplitude.
- Mean – Displays the average signal amplitude.
- Standard Deviation – Indicates the variation in signal amplitude.

- Dominant Frequency – Displays the strongest frequency component obtained from FFT.

8 Key Features

Feature	Description	Benefit
ECG Visualization	Displays heartbeat waveform	Easy interpretation
Beat Navigation	Browse beats	Dataset exploration
FFT Analysis	Frequency spectrum	Spectral analysis
Peak Detection	Detect dominant peaks	Feature extraction
Moving Average Filter	Smooths ECG	Noise reduction
Median Filter	Removes impulsive noise	Preserves waveform
Low-pass Filter	Removes HF noise	Signal enhancement
High-pass Filter	Removes baseline drift	Cleaner ECG
Statistics Panel	Numerical parameters	Quantitative analysis
Export Signal	Save CSV	Data sharing
Export Plot	Save image	Reporting

9 Project Structure

```
Interactive-Medical-Signal-Analysis-Workbench/
|
|-- callbacks/
|-- gui/
|-- plotting/
|-- processing/
|-- utils/
|-- main.sce
|-- README.md
|-- screenshots
|-- demo video
```

Folder Description

- `callbacks/` : Handles button callbacks

- gui/ : Creates interface
- plotting/ : ECG plotting
- processing/ : DSP algorithms
- utils/ : Helper functions
- main.sce : Main application
- screenshots : Pictures of the application
- demo video : Contains the demo video of the application

10 GUI Screenshots

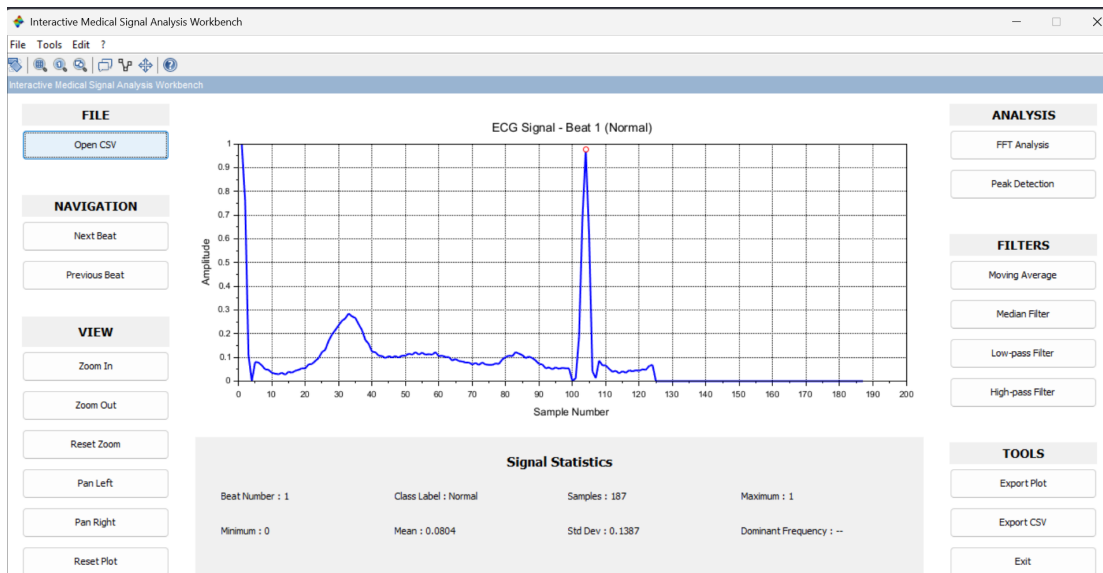


Figure 2. ECG waveform of the selected heartbeat with automatic R-peak detection highlighted.

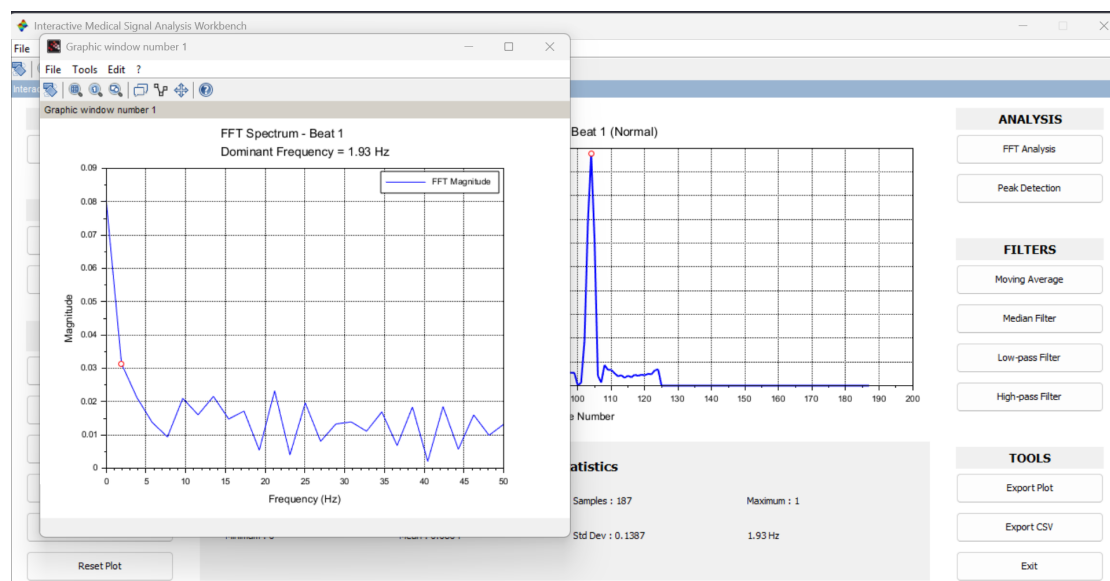


Figure 3. Frequency spectrum of the selected ECG heartbeat obtained using Fast Fourier Transform (FFT).

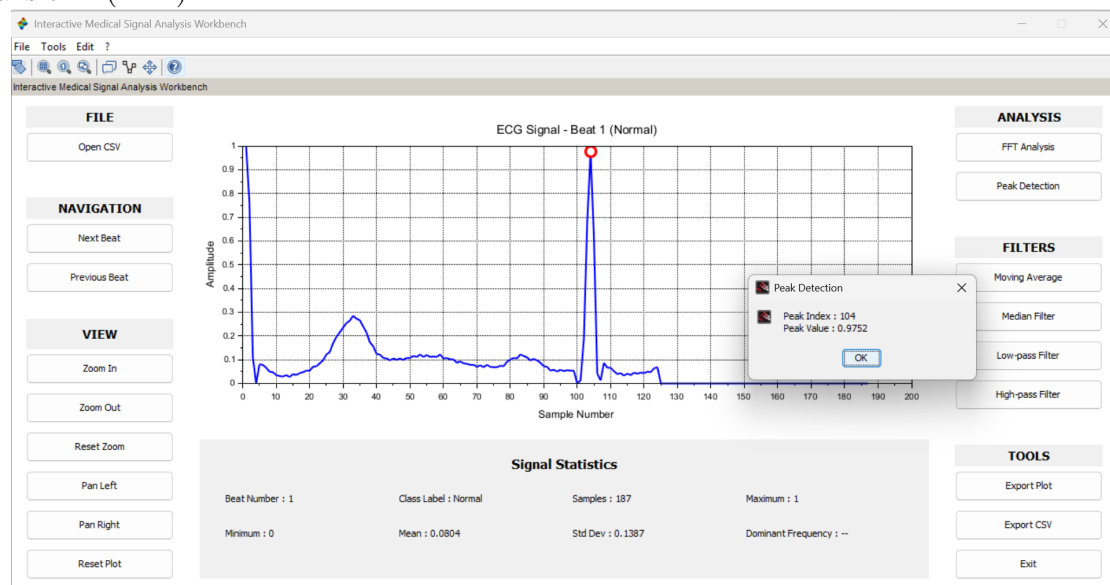


Figure 4. Peak detection result displaying the detected R-peak location and amplitude.

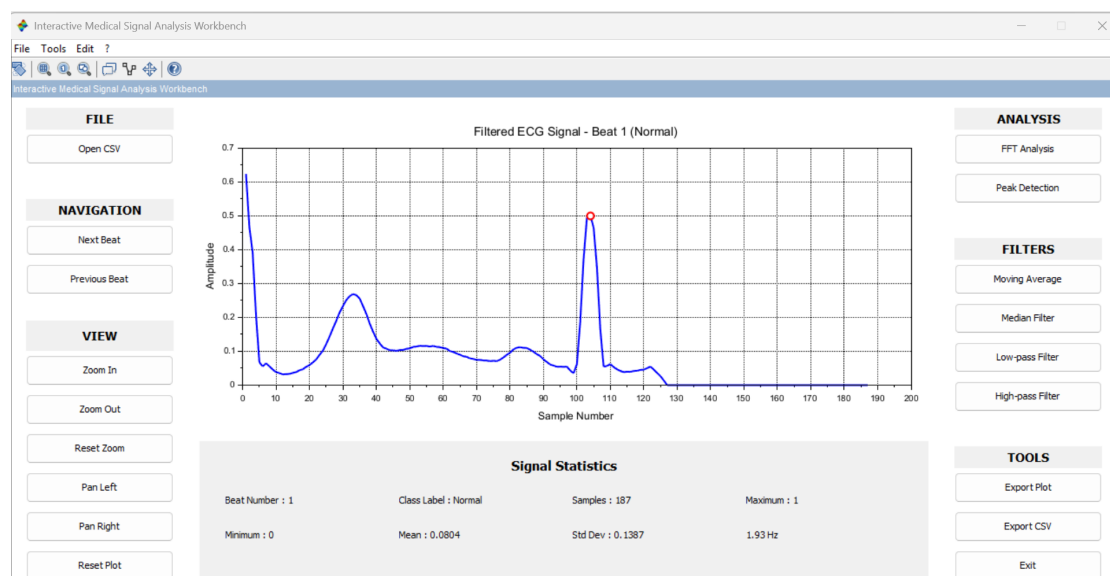


Figure 5. ECG signal after applying the Moving Average filter.

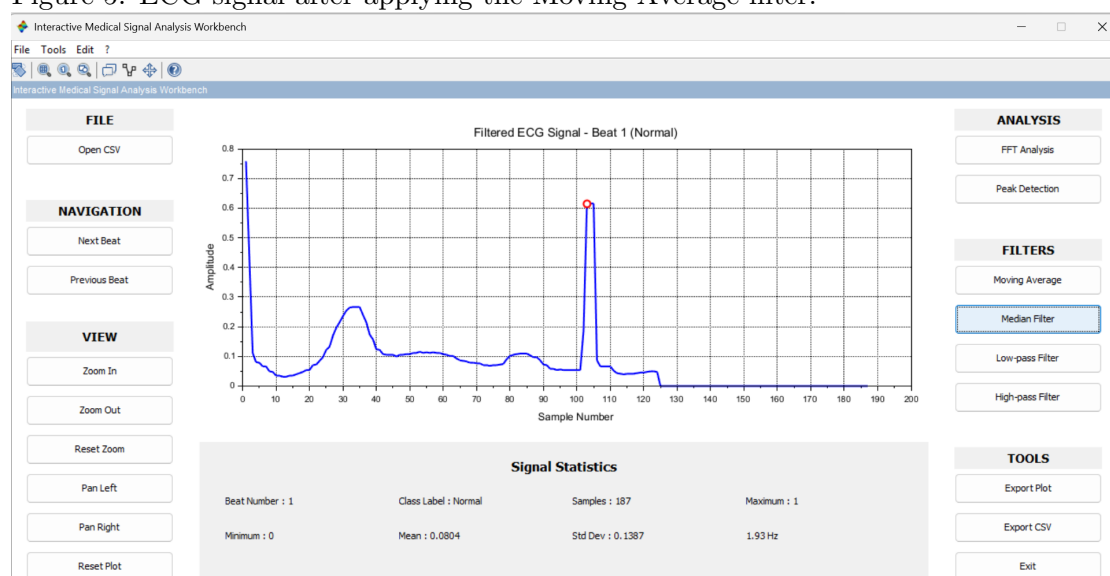


Figure 6. ECG signal after Median filtering.

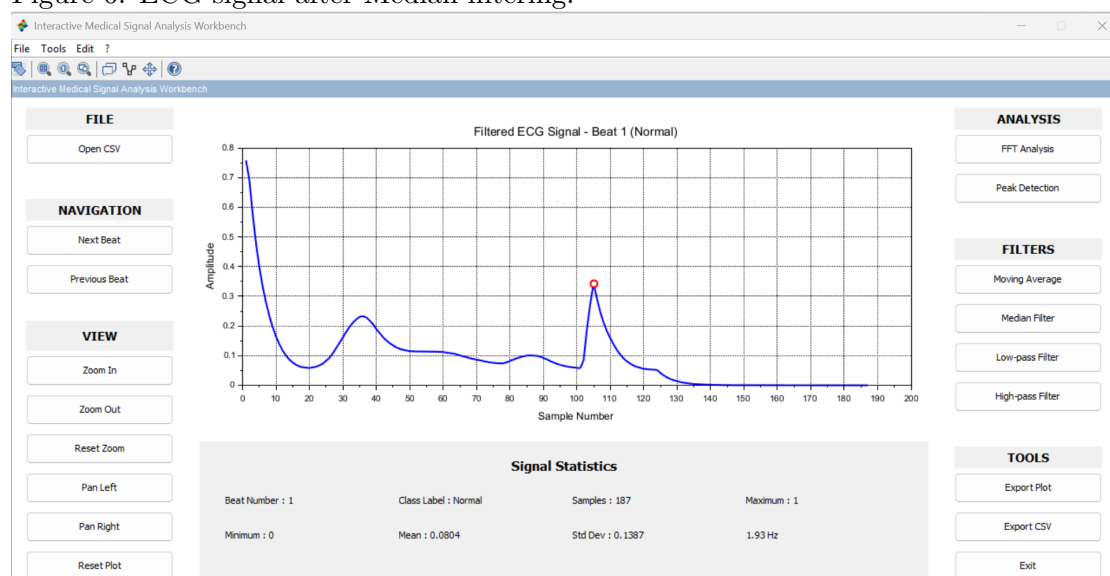


Figure 7. ECG signal after applying the Low-pass filter.

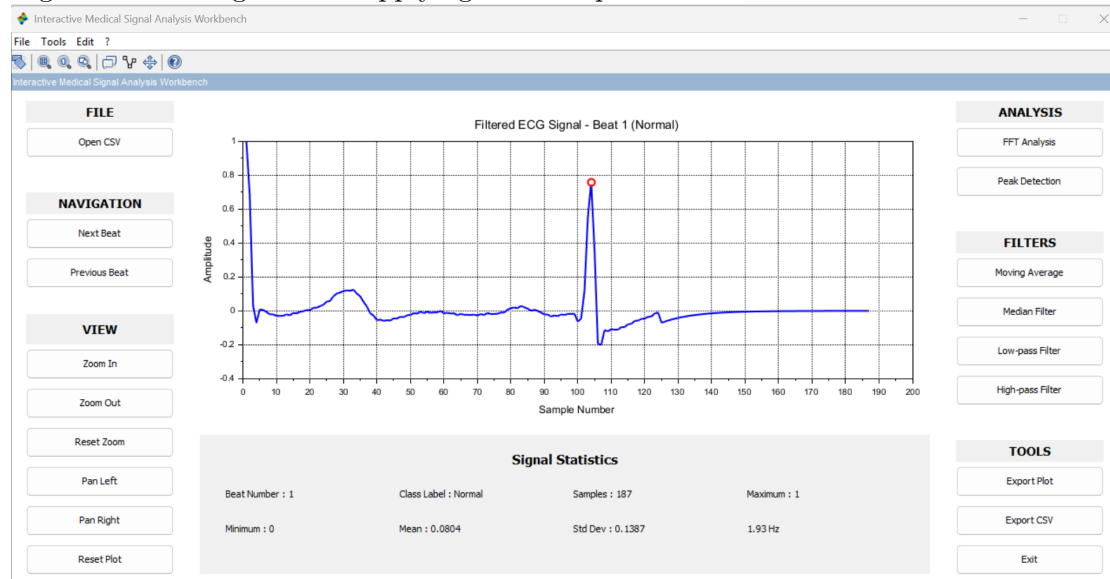


Figure 8. ECG signal after applying the High-pass filter.

11 References

1. G. B. Moody and R. G. Mark, "The MIT-BIH Arrhythmia Database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.
2. PhysioNet, "MIT-BIH Arrhythmia Database." Available: <https://physionet.org/>
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4. A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd Edition, Pearson, 2010.
5. Steven W. Smith, *The Scientist and Engineer's Guide to Digital Signal Processing*, California Technical Publishing, 1997.